Advanced Formal Tools

CTL Model Checker

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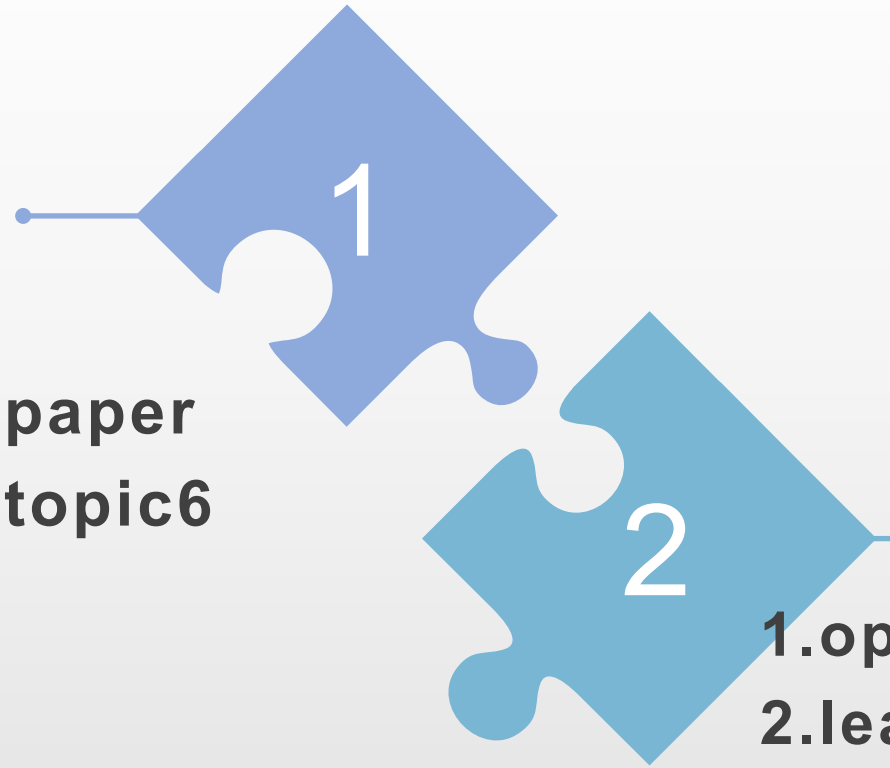
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CONTENTS

Current Status

1. Understanding of paper
2. Understanding of topic6
3. What we did



Goal

1. optimize what we did
2. learn advanced coding strategies
3. Implement our own case study

CTL model checking algorithm

Model	SMV	VIS	BINGO	
			Bwd	Fwd
gigamax	3.6sec 1.6MB	21.3sec 4.9MB	9.3sec 3.7MB	9.1sec 3.7MB
atm_sw	1734sec 8.3MB	*space*	9447sec 27.1MB	1158sec 23.1MB
dh_1	—	138.7sec 6.8MB	29.6sec 3.7MB	20.3sec 3.8MB
dh_2	—	*space*	*time*	1357sec 31.5MB
vpp	*time**	*space*	45569sec 17.5MB	223.3sec 13.2MB
pipe_s	23715sec 60.2MB	51.7sec 6.1MB	12429sec 5.5MB	64.6sec 4.9MB
pipe_d	—	*space*	*time*	8979sec 249.7MB

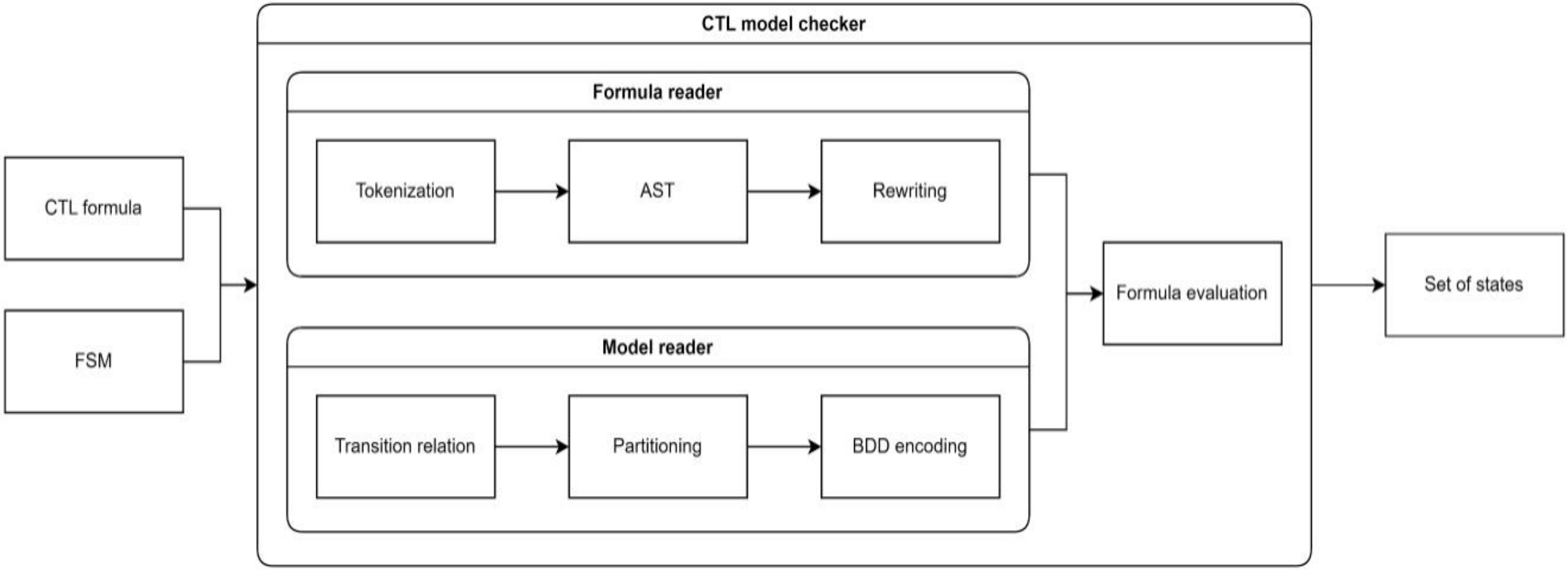
'*time*' exceeded 24 hours in CPU time.
'*space*' exceeded 300 megabytes of memory.
'*' failed while building a transition relation.

Table 3: Performance of CTL model checking

--based on forward state traversal

CTL model checker constructure

--based on forward state traversal

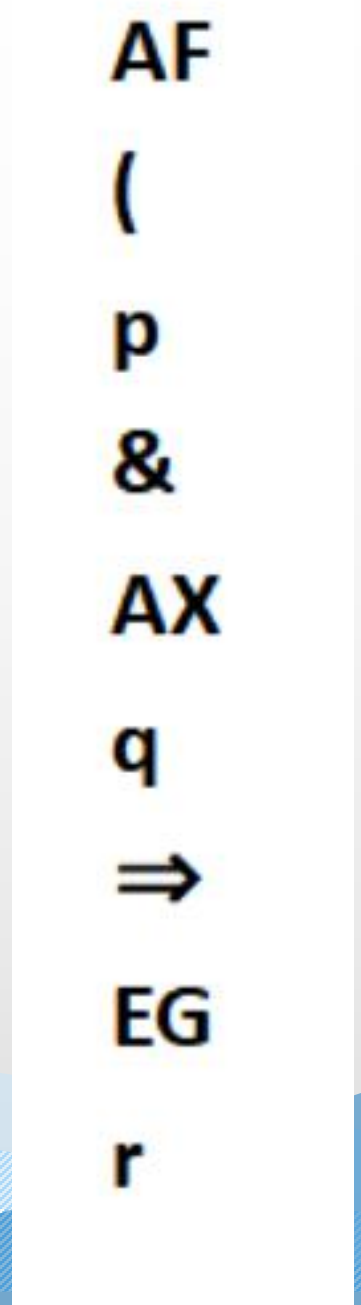


What we did

1. CTL formula to tokenizaition

example:

AF((p & AX q) ⇒ EG r)

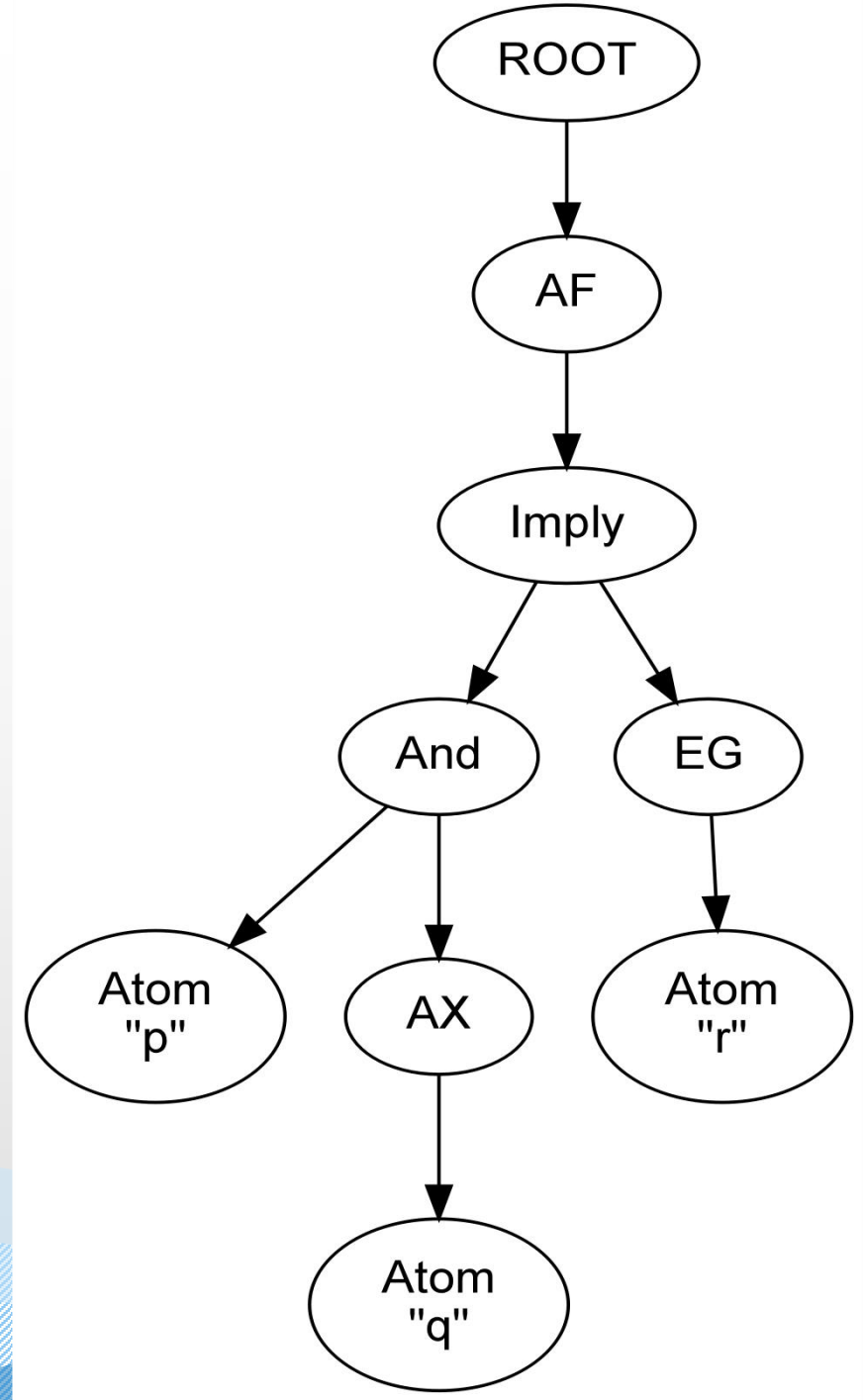


What we did

2. Tokenizaition to AST

example:

AF((p & AX q) ⇒ EG r)



What we did

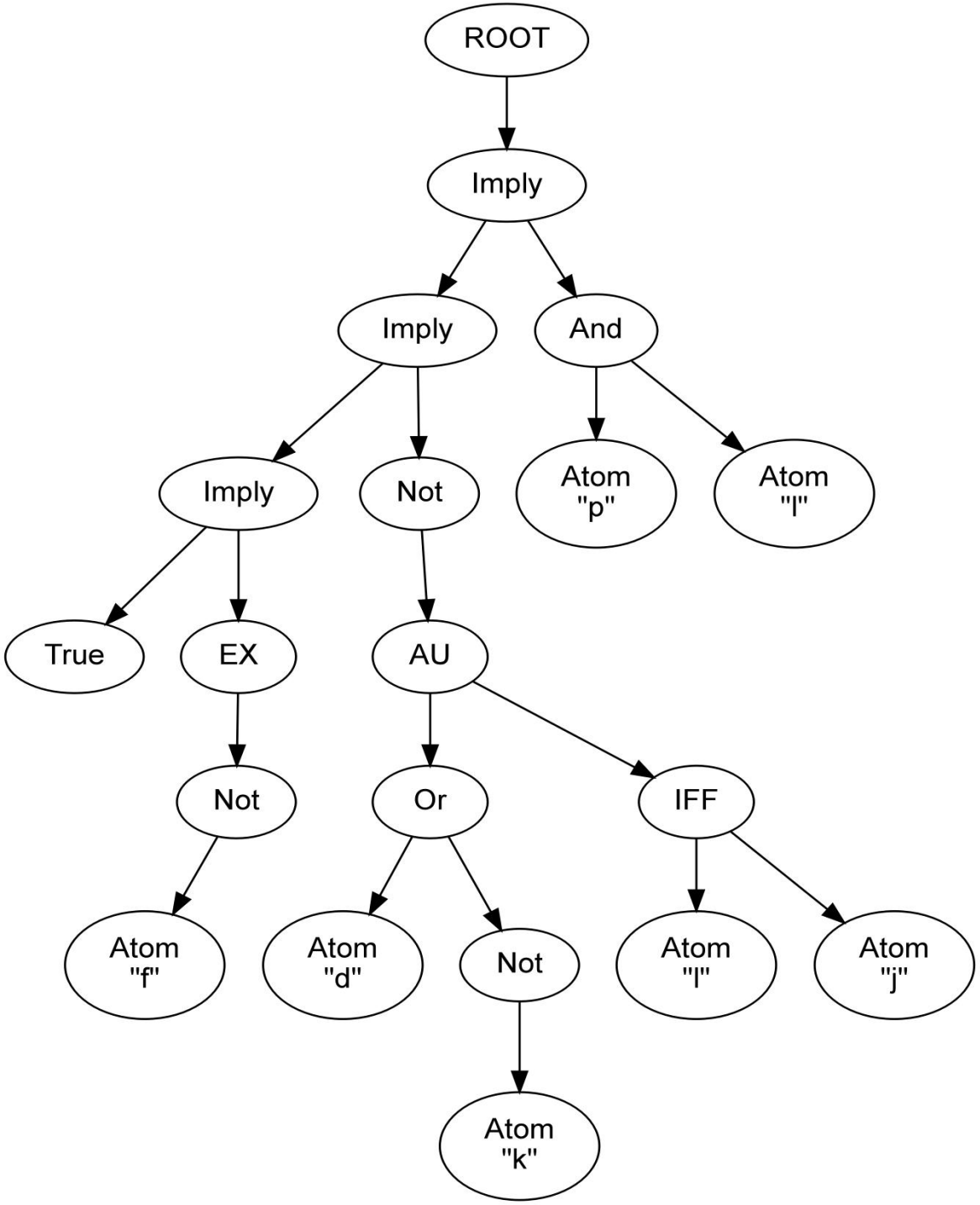
3. Another example AST

example:

((true⇒EX(¬f))

⇒(¬A[(d∨(¬k))U(l⇔j)])

⇒(AXp∧l)



What we want to do next step?

- **FIRST:**

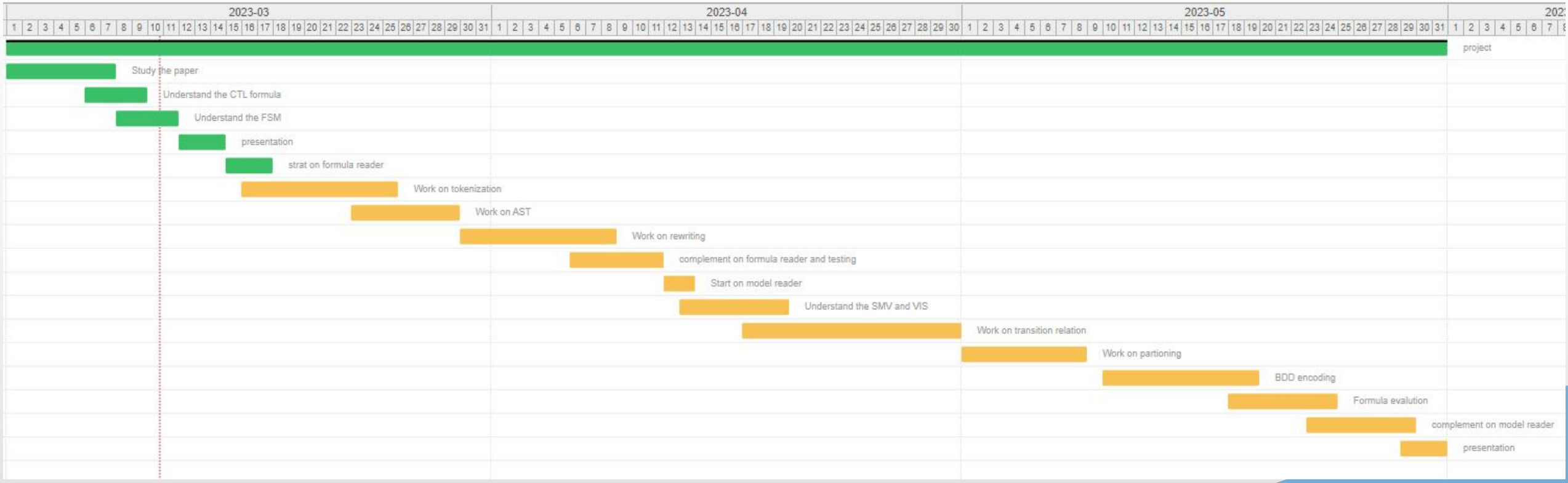
Study more about the AST and optimize AST;

Simplify complex CTL formulas by using rewriting techniques;

- **SECOND:**

Fous on the conversion procedure, try to convert all CTL temporal operators into forward traversal operators.

Schedule


2023-03
2023-04
2023-05
project
Study the paper
Understand the CTL formula
Understand the FSM
presentation
strat on formula reader
Work on tokenization
Work on AST
Work on rewriting
complement on formula reader and testing
Start on model reader
Understand the SMV and VIS
Work on transition relation
Work on partioning
BDD encoding
Formula evalution
complement on model reader
presentation

THANK YOU